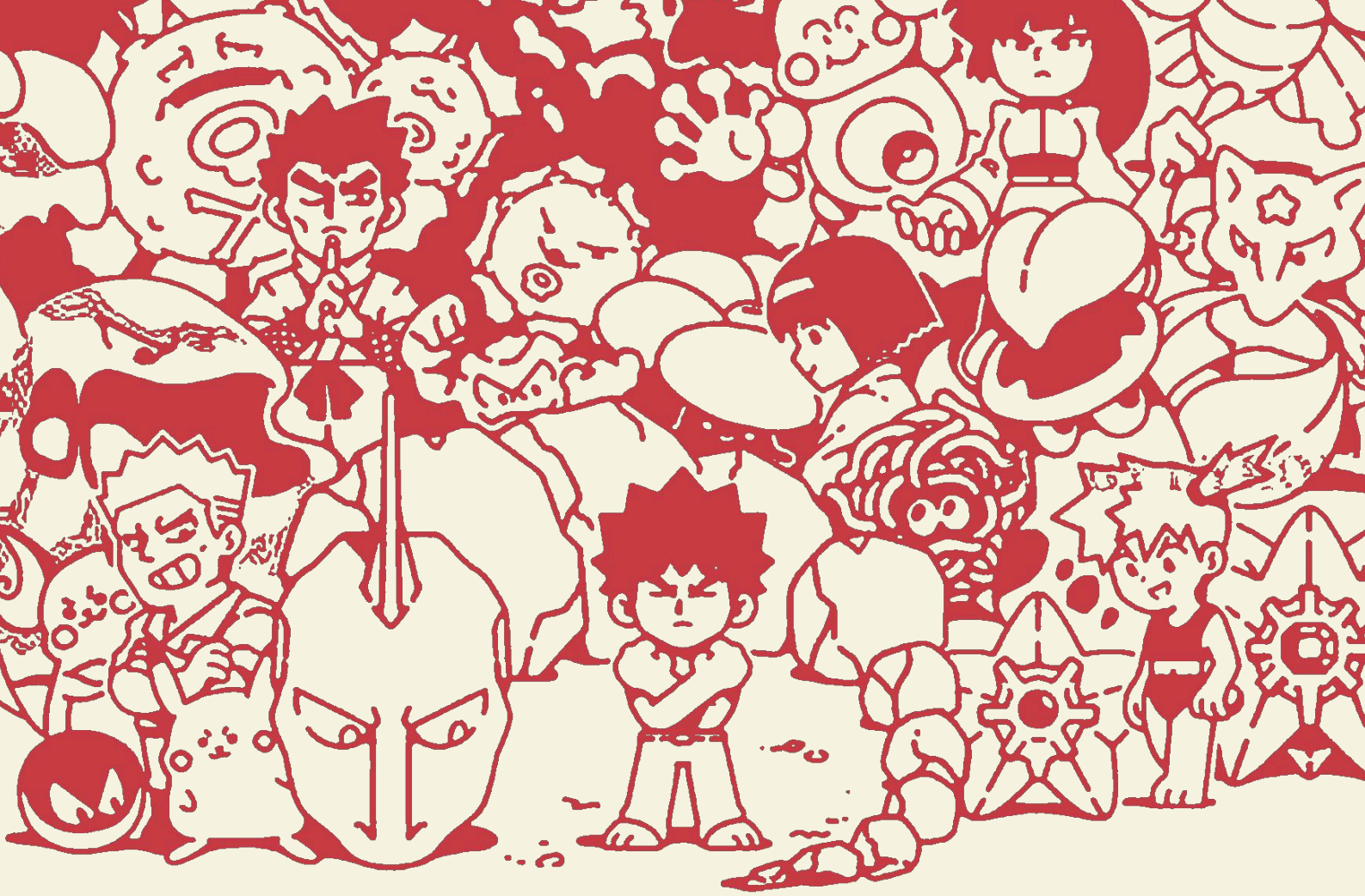

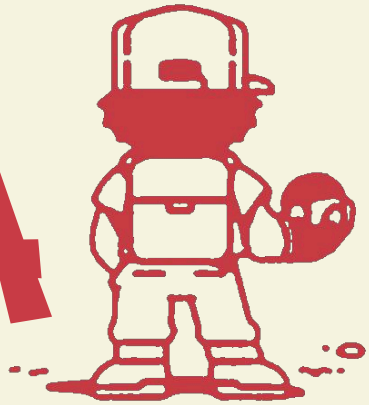

GOTTA CODE EM ALL

ODT CAPSTONE PROJECT

Test Planning and results

Sachit Patnaik
Tanay Dhavate

BITS Design
2025-29

TESTS CONDUCTED

	Method	Success Criteria
ADC Feasibility	Tested different GPIO pins and conductor materials (copper tape, jumper cables) to find the most stable ADC read	A consistent, noise-free voltage reading per figure
ADC Value Mapping	Plugged each resistor into pin 33 with jumper cables and logged average readings across 20 samples	10 distinct, non-overlapping ADC ranges , one per Pokémon
Audio File Compatibility	Tested each WAV file individually through the I2S amplifier; tried different file formats, sample rates, and bit depths	Clean, clear audio playback for all 10 Pokémon cries and result sounds
Animation Binary Files	Loaded each .bin frame file onto the OLED and verified correct rendering per Pokémon	All 10 sprites display correctly with accurate light/dark mapping
User Interaction	Handed device to multiple first-time users with no instructions and observed the full interaction	Users complete scan and catch attempt unaided within a reasonable time
System Stability	Ran repeated full cycles; identified crash points and added deliberate delays and SD card checks into the code	Device completes multiple cycles without freezing, hanging, or crashing

TESTS CONDUCTED

	Outcome
ADC Feasibility	GPIO 33 with jumper cables gave the most stable reads; copper tape was inconsistent; GPIO 12 nearly damaged the ESP32
ADC Value Mapping	10 distinct ranges confirmed with generous spacing between each value
Audio File Compatibility	Distorted audio resolved by adding a 75Ω resistor between GPIO and amp data pin; clean playback achieved across all files
Animation Binary Files	All 10 sprites render correctly; light/dark inversion manually adjusted per Pokémon
User Interaction	All users successfully completed the full interaction unaided; many returned for repeat attempts
System Stability	Crashes resolved by redoing wiring, adding boot delays, and inserting SD card initialisation checks

OBSERVATIONS

User Behaviour

• Users instinctively picked up the figures and attempted to plug them in without instruction
• The 5-second hold confused some users initially but the OLED countdown made it clear quickly
• Multiple users returned to try different figures : the variety of Pokémon and rarity system drove repeat engagement
• The physical Pokéball shaking created a strong reaction and was consistently the most memorable moment

Pain Points

• First-time users occasionally released the SCAN button too early before understanding the full 5-second hold
• Some users were unsure whether the device had registered their plug-in before the animation started
• The boot sequence was occasionally mistaken for a malfunction during the LED fade-off

#130

IMPROVEMENTS REQUIRED AFTER USER TEST

- Add a short visual or audio cue immediately on plug-in to confirm the figure has been detected before the full reveal
- Display a brief "Hold for 5s" reminder more prominently during the first catch attempt
- Consider a clearer boot complete indicator so users know the device is ready to scan
- Explore PCB fabrication to eliminate breadboard wiring instability for a more robust final build
- Add a wider ADC deadzone between figure values as a safeguard against connector wear over time